CoolJet
Refrigerant-less Cooling

SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be **University** *u/s 3 of UGC Act, 1956)*

Startup Pitch Deck

CoolJet | Refrigerant-less Cooling Solution

Presenter Name
RUDRA SHEKHAR DUBEY

Institution
SRM Institute of Science & Technology

Date

Problem Statement

CoolJet
Refrigerant-less Cooling

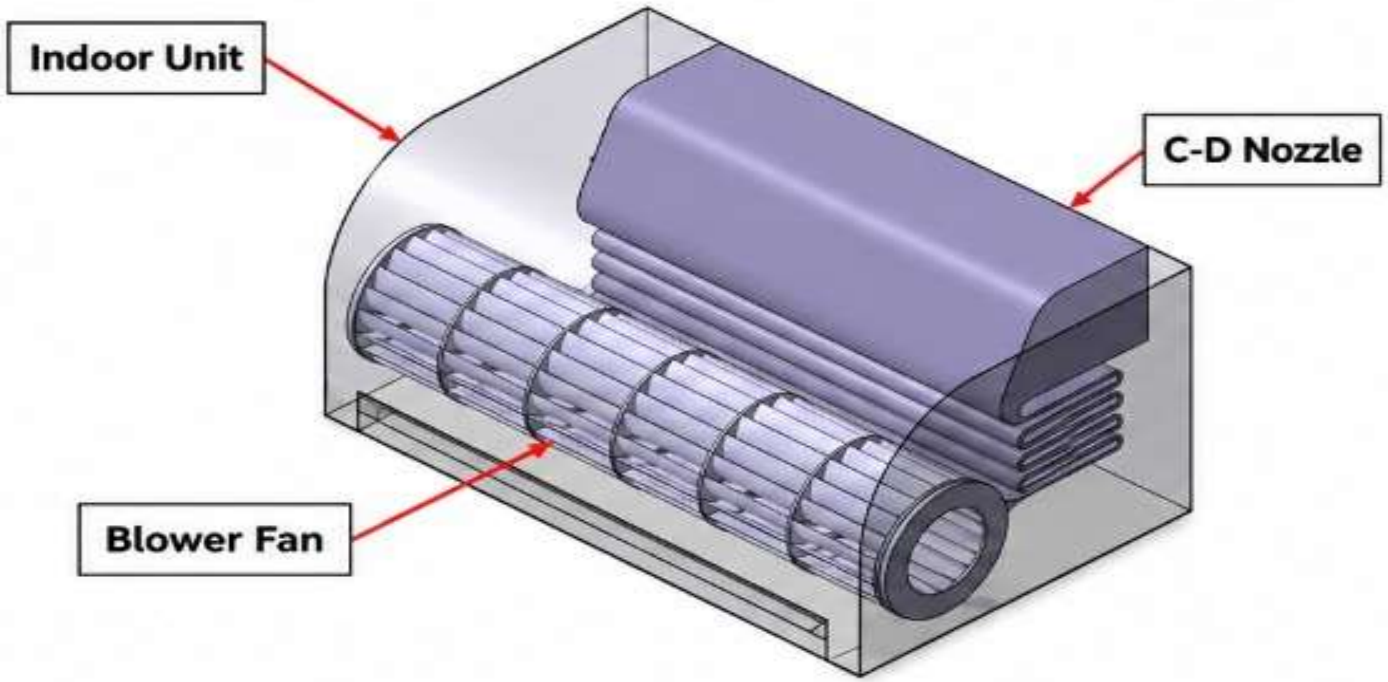
- Traditional air conditioners rely on refrigerants and compressors, which contribute to **greenhouse gas emissions, high energy consumption and complex maintenance.** These systems also pose risks of refrigerant leakage, require frequent servicing and are not environmentally sustainable.

- There is a growing demand for cooling solutions that are **energy-efficient, eco-friendly, safe and easy to maintain**—especially for homes, offices, healthcare facilities, rural areas and defense applications where reliability and sustainability are critical.

- Current cooling technologies fail to address the need for a **refrigerant-less system** that delivers effective cooling with **low power consumption, minimal maintenance and long-term environmental benefits.**

Solution / Product Picture

CoolJet
Refrigerant-less Cooling



- Our solution is a Refrigerant-less Air Conditioner that uses Convergent–Divergent (C–D) nozzles to cool air without any refrigerants or compressors.
- Ambient air is accelerated through the C–D nozzle; at the throat its pressure drops and temperature decreases (adiabatic expansion) to deliver cool air.
- The system is simple, energy-efficient, eco-friendly and uses only a blower fan, no moving parts in the nozzle, and low maintenance.
- Innovation: Adapting aerospace nozzle technology for sustainable, cost-effective and safe cooling—eliminating refrigerants and reducing power consumption.

3D DESIGN OF THE PROJECT

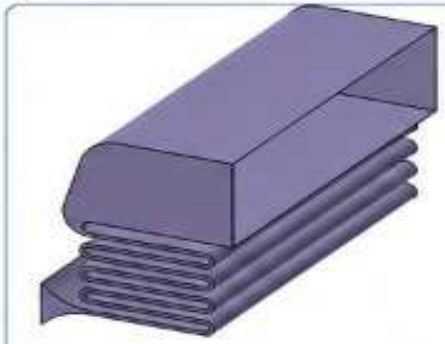


Fig 5.1: Front Side 3D View of C-D Nozzle

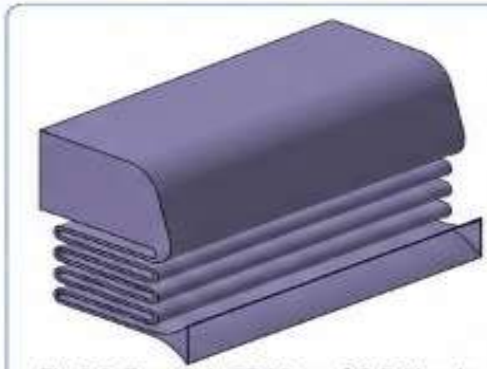


Fig 5.2: Rear Side 3D View of C-D Nozzle

DIMENSION FOR CONVERGENT DIVERGENT NOZZLE

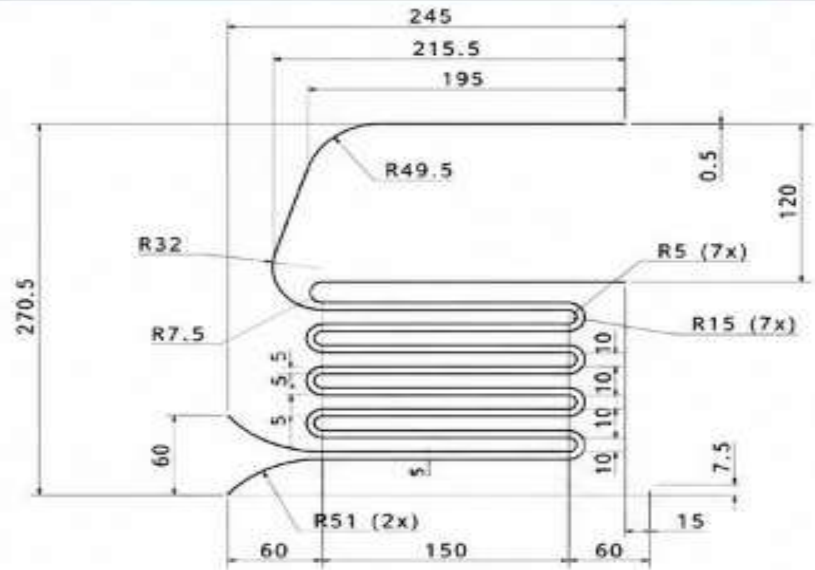


Fig 5.3: Dimension for Convergent Divergent Nozzle

Market Opportunity

CoolJet
Refrigerant-less Cooling

TARGET APPLICATION AREAS

Homes, apartments, hostels, schools, offices, hospitals, small businesses, rural & off-grid areas, and defense & field applications.

MARKET SIZE & GROWTH POTENTIAL

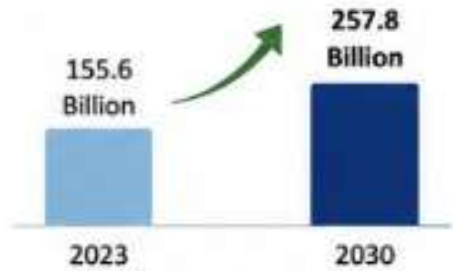
The global air conditioning market was valued at USD 155.6 billion in 2023 and is projected to reach USD 257.8 billion by 2030, growing at a CAGR of 7.5%. India's AC market is expected to grow from ~USD 9.3 billion (2023) to ~USD 14.5 billion by 2030 (CAGR ~6.6%).

KEY MARKET DRIVERS

Rising demand for energy-efficient & sustainable cooling, phase-down of HFC refrigerants, increasing electricity costs, government push for green technologies, and growing adoption in hot & humid climates create a strong opportunity.

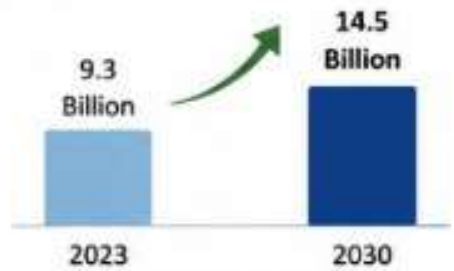
MARKET AT A GLANCE

GLOBAL AC MARKET SIZE



CAGR 7.5%

INDIA AC MARKET SIZE



CAGR 6.6%

TARGET CUSTOMER SEGMENTS

- Homes & Apartments
- Offices & Workspaces
- Schools & Colleges
- Rural & Off-Grid Areas
- Hospitals & Clinics
- Defense & Field Applications
- Small Businesses

KEY TRENDS

- Shift towards refrigerant-free & eco-friendly solutions
- Rising energy costs & demand for energy-efficient cooling
- Phase-down of HFC refrigerants (global regulations)
- Government push for green technologies
- Increasing adoption in hot & humid climates

Huge opportunity for a Refrigerant-less Air Conditioner using C-D Nozzles – sustainable, low-power, and future-ready solution.

Technology / Innovation

CoolJet
Refrigerant-less Cooling

TECHNOLOGY OVERVIEW

CoolJet leverages aerospace-inspired Convergent–Divergent (C–D) nozzles with a blower fan. Ambient air is accelerated through the nozzle; at the throat, pressure drops and temperature reduces (adiabatic expansion), delivering cool air at the outlet without any refrigerants or compressor.

INTELLECTUAL PROPERTY & R&D FOCUS

Proprietary C–D nozzle geometry and system integration optimized through CFD analysis. Novel application of nozzle technology for room cooling. Patent filing in progress.

TECHNOLOGY READINESS

TRL Level 5 – Prototype validated in relevant environment. CFD validated design with optimized nozzle. Prototype assembled and tested for performance.

TRL 5

TECHNOLOGY READINESS LEVEL 5

Prototype validated in relevant environment

TECHNOLOGY IN ACTION



Pressure drop at the throat causes temperature drop (adiabatic expansion) resulting in cooled air.

KEY INNOVATIONS

AEROSPACE C–D NOZZLE TECHNOLOGY

Aerospace-inspired adiabatic expansion for cooling

REFRIGERANT-LESS & ECO-FRIENDLY

No harmful refrigerants, zero leakage, environment safe

ENERGY EFFICIENT

Operates with only a blower fan, low power consumption

SIMPLE & LOW MAINTENANCE

Few moving parts, no compressor, low maintenance

VALIDATED DESIGN

CFD optimized & prototype tested for performance

Business Model

CoolJet
Refrigerant-less Cooling

REVENUE GENERATION

CoolJet will generate revenue by designing, manufacturing and supplying refrigerant-less air conditioning units and components to residential, commercial and industrial customers.

REVENUE MODEL

Product Sales
One-time sale of CoolJet AC units and C-D nozzle kits.

Maintenance & AMC
Annual Maintenance Contracts and paid servicing.

Component Supply
Supply of C-D nozzles and retrofitting kits to OEMs and HVAC players.

Licensing & Partnerships
Technology licensing and collaboration with HVAC manufacturers.

CUSTOMER ACQUISITION STRATEGY

- Direct Sales Team
- Industry Exhibitions & Events
- Digital Marketing & Online Presence
- Channel Partners & Distributors
- Institutional & Government Tenders

TARGET CUSTOMER SEGMENTS

- Homes & Apartments
- Offices & Commercial Buildings
- Industrial Facilities
- Schools, Colleges & Hospitals
- Defence & Field Applications

VALUE PROPOSITION

- Refrigerant-less & Eco-friendly
- Energy Efficient & Low Power Consumption
- Low Maintenance & Simple Design
- Cost Effective Cooling Solution
- Sustainable & Future Ready

REVENUE STREAMS

Product Sales → Maintenance & AMC → Component Supply → Licensing & Partnerships

Affordable, sustainable and scalable cooling solution creating long-term value for customers and partners.

Go-to-Market (GTM) Strategy

CoolJet
Refrigerant-less Cooling

MARKET ENTRY APPROACH

We will target residential, commercial and industrial customers through a phased market entry focusing on pilots, partnerships and scaling across key channels.

GO-TO-MARKET STRATEGY

Direct Sales & B2B Outreach
Dedicated sales team for corporate, institutional and industrial clients.

Strategic Partnerships
Collaborate with HVAC OEMs, distributors, and system integrators.

Digital Marketing
SEO, social media, technical content, webinars and product demos.

Pilot Projects & Case Studies
Deploy pilots in key segments to build credibility and showcase performance.

Exhibitions & Industry Events
Participate in HVAC, sustainability and innovation events.

ROLL-OUT TIMELINE

Our phased roll-out ensures validated performance, strong partnerships and scalable growth.

TARGET CUSTOMER SEGMENTS

- Homes & Apartments
- Offices & Commercial Buildings
- Industrial Facilities
- Schools, Colleges & Hospitals
- Defense & Field Applications

MARKETING & SALES STRATEGY

Identify & Engage Target Customers → Demonstration & Pilot Projects → Case Studies & References → Commercial Proposal → Close Deals & Scale Up

PHASED ROLL-OUT PLAN (3–6–12 MONTHS)

0–3 MONTHS

- Finalize product & pilot units
- Build brand presence & digital channels
- Identify early adopter customers
- Initiate pilot projects in select segments

3–6 MONTHS

- Complete pilot deployments
- Collect performance data & feedback
- Develop case studies & testimonials
- Expand partnerships & channel network

6–12 MONTHS

- Scale production & supply chain
- Enter new markets & geographies
- Increase sales & brand visibility
- Achieve recurring orders & AMC base

Our GTM approach ensures strong market entry, validated performance, and scalable growth across high-potential segments.

Impact

CoolJet
Refrigerant-less Cooling

ENVIRONMENTAL & CLIMATE BENEFITS

CoolJet provides a sustainable, energy-efficient and cost-effective cooling solution that reduces dependence on harmful refrigerants, lowers electricity consumption and improves quality of life in homes, workplaces and communities.

SOCIAL & COMMUNITY BENEFITS

Our solution supports global sustainable development goals and national missions by driving energy efficiency, climate action and green technology adoption while enhancing well-being and creating opportunities.



ECONOMIC & VALUE CREATION

Beyond financial returns, CoolJet contributes to a cleaner environment, healthier communities, energy security and a sustainable future for generations to come.

ENVIRONMENTAL IMPACT

Lower Emissions
No refrigerants, zero ODP & ultra-low GWP.

Energy Efficiency
Up to 30–40% less energy consumption compared to conventional ACs.

Climate Action
Reduces greenhouse gas emissions and supports a low-carbon future.

Resource Conservation
Minimal use of resources and low maintenance reduces waste.

No Leakage Risk
Eliminates refrigerant leakage, protecting the ozone layer and ecosystems.

SOCIAL IMPACT

Better Comfort
Provides affordable and reliable cooling for better living and productivity.

Health & Safety
Improves indoor air quality with no toxic refrigerants or leakage risks.

Education & Awareness
Promotes awareness of sustainable technology in communities.

Job Creation
Creates opportunities in manufacturing, installation and servicing.

Energy Security
Reduces power demand and supports national energy security.

ECONOMIC IMPACT

Lower Operating Cost
Reduced electricity bills and low maintenance cost.

Affordable Solution
Lower upfront and lifecycle cost compared to conventional ACs.

Make in India
Encourages local manufacturing and self-reliance.

Boosts Green Economy
Supports sustainable businesses and green technology adoption.

Long-term Value
Delivers durable value for customers, society and the environment.

CoolJet creates a positive impact across environment, society and economy—building a cooler, cleaner and sustainable world beyond profit.

Development Roadmap & Resource Plan

CoolJet
Refrigerant-less Cooling

A phased approach to ensure systematic development, market validation and scalable growth.

PHASE 1
0–3 MONTHS

FOUNDATION & VALIDATION

- Finalize design & simulations
- Build and test functional prototype
- Performance validation
- Initial pilot installations

PHASE 2
3–6 MONTHS

PILOT & MARKET ENTRY

- Deploy pilot projects
- Collect performance data & feedback
- Refine design for scalability
- Secure early customers & MoUs

PHASE 3
6–9 MONTHS

SCALE & PARTNERSHIPS

- Scale up manufacturing process
- Build supply chain & vendor network
- Expand partnerships with HVAC OEMs
- Broaden market outreach

PHASE 4
9–12 MONTHS

COMMERCIAL LAUNCH

- Full-scale product launch
- Enter new markets & geographies
- Strengthen brand & marketing
- Establish AMC & service network

RESOURCE ALLOCATION (ESTIMATED)

RESOURCE CATEGORY	0–3 MONTHS (%)	3–6 MONTHS (%)	6–9 MONTHS (%)	9–12 MONTHS (%)	TOTAL (%)
R&D & Product Development	40	25	15	10	**90**
Manufacturing & Operations	10	20	30	20	**80**
Marketing & Sales	10	20	25	30	**85**
Partnerships & Business Development	10	15	20	20	**65**
Testing, Certification & Compliance	10	10	10	10	**40**
TOTAL	**80**	**90**	**100**	**90**	**360**

This phased approach ensures **efficient resource utilization, mitigates risk and drives scalable growth of CoolJet.**

Financial Overview

CoolJet
Refrigerant-less Cooling

FINANCIAL OUTLOOK

This 3-year projection is based on pilot deployments, market expansion and gradual scale-up of manufacturing and sales.

CAPITAL REQUIREMENT

Seeking funding to scale production, expand market reach and strengthen technology & team.

FINANCIAL DISCIPLINE

Conservative assumptions with a focus on profitability from Year 3 onwards.

FUNDING REQUIRED

₹ 75,00,000

Pre-Seed Round

USE OF FUNDS



- Product Development & R&D (30%)
- Manufacturing & Prototyping (25%)
- Marketing & Sales (20%)
- Team & Operations (15%)
- Contingency (10%)

This funding will enable us to accelerate product readiness, validate in larger markets and scale growth.

3-YEAR FINANCIAL PROJECTION (INR)

Particulars	Year 1 (₹)	Year 2 (₹)	Year 3 (₹)
Revenue from Product Sales	75,00,000	2,25,00,000	5,25,00,000
Revenue from AMC & Services	15,00,000	45,00,000	1,05,00,000
Total Revenue	**90,00,000**	**2,70,00,000**	**6,30,00,000**
Cost of Goods Sold (COGS)	40,00,000	1,20,00,000	2,50,00,000
Gross Profit	**50,00,000**	**1,50,00,000**	**3,80,00,000**
Operating Expenses	60,00,000	1,20,00,000	1,80,00,000
EBITDA	**-10,00,000**	**30,00,000**	**2,00,00,000**
Profit Before Tax	-15,00,000	20,00,000	1,70,00,000
Net Profit	**-15,00,000**	**16,00,000**	**1,30,00,000**
EBITDA Margin (%)	-11%	11%	32%

KEY ASSUMPTIONS

- Conservative sales ramp-up based on pilot-to-market approach.
- Gross margin improves with scale and supply chain optimization.
- Operating leverage drives profitability from Year 2 onwards.
- Focus on high-impact segments with recurring revenue (AMC).
- Long-term value creation through sustainable and scalable solutions

CoolJet is building a **profitable, scalable** and **sustainable** business with strong long-term potential.

Budget Split up

CoolJet
Refrigerant-less Cooling

Knowledge Partner:			
Project Milestone / Activity	Total Estimated Cost (in Rupees)	IVP Contribution (in Rupees)	Applicant Contribution (in Rupees)
1) R&D / Design Studies	50,000	45,000	5,000
2) Market Research	40,000	36,000	4,000
3) Material Studies / Raw materials	45,000	40,500	4,500
4) Fabrication / Manufacturing charges for PoC & Prototype Development	1,30,000	1,17,000	13,000
5) Testing and Validation	55,000	49,500	5,500
6) Feasibility Studies	30,000	27,000	3,000
7) Intellectual Property Rights / Patent Filing	25,000	22,500	2,500
8) Any other expenses (to be listed)	20,000	18,000	2,000
Total Estimated Cost (in Rupees)	**3,25,000**	**3,00,000**	**25,000**
IVP Contribution (in Rupees)	–	3,00,000	–
Applicant Contribution (in Rupees)	–	–	**25,000**

Total Estimated Cost: **₹ 3,25,000** | IVP Contribution (90% up to ₹ 3,00,000): **₹ 3,00,000** | Applicant Contribution: **₹ 25,000**

Note: As per scheme guidelines, IVP contribution is restricted to 90% of the project cost subject to a maximum of ₹ 3,00,000.